Gaia DR3 4277855016732107520: A High-Priority Spectroscopic Follow-Up Target from the Gaia DR3 Astrometric Orbital Catalogue

Joel Ayala-Baez[1]⋆
[1] *Independent Researcher*

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ABSTRACT

We identify Gaia DR3 4277855016732107520 as a high-priority spectroscopic follow-up target from the Gaia DR3 non-single-star catalogue. The source carries a purely astrometric `Orbital` solution with no independent radial-velocity orbit; all inferred companion properties are conditional on the Gaia pipeline model, and $\gtrsim$ 30–50 per cent of comparable solutions fail independent RV validation.

Adopting $M_1 = 1.34 \pm 0.40\,\mathrm{M}_\odot$ (GSP-Phot isochrone matching, cross-checked against a PARSEC grid; RUWE = 9.31) and a $\times 1.7$ parallax-error inflation, a Monte Carlo posterior yields median $M_2 = 12.3\,\mathrm{M}_\odot$ (90 per cent CI $[5.0, 35.9]\,\mathrm{M}_\odot$; $P(M_2 > 5\,\mathrm{M}_\odot) \approx 95$ per cent conditional on the solution being genuine). Synthetic-photometry companion exclusion rules out a main-sequence star at the inferred mass, while a stripped-helium-star scan over 1–15 $\mathrm{M}_\odot$ finds no photometrically hidden solution.

The predicted $K_1 \approx 65\,\mathrm{km\,s^{-1}}$ makes ground-based RV confirmation efficient; the source ranks among the top bright ($G < 12$) `Orbital` BH candidates by K_1. Until RV confirmation is obtained, the companion mass and nature remain entirely provisional.

Key words: binaries: visual – astrometry – stars: fundamental parameters – stars: individual: Gaia DR3 4277855016732107520

1 INTRODUCTION

Theoretical models predict 10^8–10^9 stellar-mass black holes (BHs) in the Milky Way (Agol & Kamionkowski 2002; Shapiro & Teukolsky 1983), yet fewer than $\sim$70 are dynamically confirmed, nearly all in X-ray binaries undergoing active accretion (Remillard & McClintock 2006). The vast majority are expected to reside in non-interacting (dormant) systems, producing no electromagnetic signature beyond gravitational influence on a luminous companion.

Gaia Data Release 3 (DR3; Gaia Collaboration et al. 2023a) has transformed the search for dormant BHs. The non-single-star (NSS) catalogue (Gaia Collaboration et al. 2023b) provides orbital solutions for $\sim$180 000 binaries, including astrometric orbits that — under the modelling assumptions of the pipeline — yield direct companion mass estimates rather than the minimum masses available from spectroscopic (SB1) solutions alone. These solutions come in several model types, ranging from the purely astrometric `Orbital` class (photocentre orbit only) to `AstroSpectroSB1` solutions that combine astrometry with an independent spectroscopic orbit. The latter provide substantially stronger constraints because the radial-velocity orbit independently confirms the period, eccentricity, and semi-amplitude. By contrast, an `Orbital` solution derives the companion mass entirely from the along-scan astrometric residuals and is therefore more vulnerable to pipeline modelling errors and scan-geometry biases.

The landmark discoveries of Gaia BH1 ($M_2 \approx 9.6\,\mathrm{M}_\odot$; El-Badry et al. 2023a), Gaia BH2 ($M_2 \approx 8.9\,\mathrm{M}_\odot$; El-Badry et al. 2023b), and Gaia BH3 ($M_2 \approx 33\,\mathrm{M}_\odot$; Gaia Collaboration et al. 2024) have established the efficacy of astrometric BH detection. Crucially, each of these three confirmed systems benefited from independent multi-epoch radial-velocity follow-up that validated the astrometric orbit — a step that has not yet been performed for the present source. Systematic searches have since expanded the candidate list (Jayasinghe et al. 2021; Thompson et al. 2019; Shenar et al. 2022; Mahy et al. 2022), but El-Badry et al. (2024) have demonstrated that a substantial fraction of high-significance `Orbital` solutions in the NSS catalogue fail independent RV checks, underscoring the need for spectroscopic confirmation of each individual candidate.

In this paper we present a single-object archival characterization and follow-up prioritization of Gaia DR3 4277855016732107520, selected from the Gaia DR3 NSS catalogue on the basis of its formally high companion mass

⋆ E-mail: jayalabaez@gmail.com

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($M_2 \simeq 12.3\,\mathrm{M}_\odot$ under pipeline assumptions), bright apparent magnitude ($G = 11.25$), and favourable observing geometry. The system is a wide binary ($P = 424\,\mathrm{d}$) with a dark companion whose mass from the astrometric orbit is $M_2 \simeq 12.3\,\mathrm{M}_\odot$ under Gaia NSS modelling assumptions. Located at ~ 657 pc in Ophiuchus ($l = 32.1°$, $b = 9.3°$), the system has moderate extinction, a parallax with ~ 10 per cent relative uncertainty, and is accessible from both hemispheres.

This source is a particularly attractive follow-up target for three reasons. First, the predicted RV semi-amplitude ($K_1 \approx 65\,\mathrm{km\,s^{-1}}$) ranks 7th out of 47 `Orbital` BH candidates and 1st among those brighter than $G = 12$ (Section 6.3), making confirmation efficient with 2–4 m class echelle spectrographs. Second, the bright apparent magnitude ($G = 11.25$) and moderate extinction keep the required exposure times short. Third, the equatorial declination (Dec $= +3.5°$) makes the source accessible from both hemispheres, with optimal visibility during May–September (Section 6.6). These observational advantages motivate presenting this source as a concrete observing-proposal target rather than waiting for Gaia DR4.

The specific contribution of this paper is a single-object follow-up dossier: (i) the first comprehensive multi-wavelength compilation for this source (Gaia, 2MASS, WISE, GALEX, ROSAT, VLASS, RACS); (ii) a Monte Carlo companion-mass posterior that propagates all identified uncertainties, explicitly flagged as a conditional working estimate; (iii) a systematic evaluation of seven non-BH scenarios with conditional verdicts; and (iv) quantitative follow-up predictions (RV semi-amplitude, epoch count, telescope class) usable for proposal planning. We also provide a quantitative comparison against the broader `Orbital` BH-candidate population (Section 6.3) to justify highlighting this particular source.

We stress from the outset that the solution type is `Orbital` (astrometry only): *no independent spectroscopic orbit exists.* A full statement of the implications appears in the standing caveat (Section 3); all inferences below are conditional on the Gaia pipeline model.

The paper is structured as follows. Section 2 describes the observational data. Section 3 presents the analysis methodology. Section 4 reports results. Section 5 evaluates non-BH explanations, conditioned on the adopted Gaia solution. Section 6 discusses implications, the evidence hierarchy, the reliability of the Gaia `Orbital` solution, and follow-up strategy, and Section 7 summarises our findings.

2 DATA

2.1 Gaia DR3 astrometry and NSS orbital solution

Gaia DR3 4277855016732107520 appears in both the main astrometric catalogue and the NSS orbital supplement with a purely astrometric orbital solution (model type: `Orbital`). This is a critical distinction: unlike `AstroSpectroSB1` solutions (which combine astrometry with a spectroscopic orbit), the `Orbital` model derives the companion mass solely from the photo-centre wobble, making it more susceptible to scan-geometry artefacts and modelling systematics.

Key astrometric parameters include parallax $\varpi = 1.5228 \pm 0.1549$ mas (distance $d \approx 657$ pc; 10.2 per cent relative error), and photometry $G = 11.25$, $G_{\rm BP} = 11.66$, $G_{\rm RP} = 10.66$ ($G_{\rm BP} - G_{\rm RP} = 0.993$). The renormalised unit-weight error RUWE $= 9.31$ is extremely elevated, as expected for an unresolved binary with large photo-centre motion — but also a regime where Gaia GSP-Phot stellar parameters become unreliable (Andrae et al. 2023). The astrometric excess noise significance is 3753σ.

The NSS orbital solution yields: period $P = 424.403 \pm 1.159$ d, eccentricity $e = 0.3427 \pm 0.0194$, and solution significance 75.4σ. Because no full NSS covariance matrix is published, we are unable to assess parameter correlations directly; our Monte Carlo approach (Section 3.6) independently samples each parameter, which may underestimate the true uncertainty if correlations are significant.

The Gaia DR3 GSP-Phot pipeline provides stellar parameters: $T_{\rm eff} = 5922$ K and $\log g = 2.93$. We caution that at RUWE $= 9.31$, GSP-Phot parameters may be systematically biased by the unresolved orbital motion; the photometric evolved-star classification should therefore be regarded as a *working assumption* pending independent spectroscopic confirmation (Section 2.4). The systemic radial velocity is $v_{\rm sys} = -29.49 \pm 0.57\,\mathrm{km\,s^{-1}}$.

2.2 Multi-wavelength photometry

We compile multi-band photometry from 2MASS ($J = 9.84$, $H = 9.42$, $K_s = 9.33$; Cutri et al. 2003) and AllWISE ($W1 = 9.23$, $W2 = 9.27$; Cutri et al. 2014). WISE colours ($W1 - W2 = -0.04$) show no mid-IR excess, consistent with a stellar photosphere. GALEX (Bianchi et al. 2017) provides a near-UV detection (NUV $= 17.52$) but no far-UV counterpart, consistent with chromospheric emission from an evolved star without a hot companion.

2.3 X-ray and radio archival data

We query the ROSAT All-Sky Survey (2RXS; Boller et al. 2016) and XMM-Newton serendipitous catalogue (4XMM-DR13; Webb et al. 2020) within 30″ and 15″ respectively, finding no counterpart. The ROSAT non-detection implies an order-of-magnitude archival upper limit of $L_X \lesssim 5 \times 10^{30}\,\mathrm{erg\,s^{-1}}$, based on the typical RASS sensitivity at the source position and distance, assuming a representative soft X-ray spectrum and negligible additional absorption correction. This limit is consistent with a non-accreting system.

At radio wavelengths, we find no counterpart in the VLA Sky Survey (VLASS; Lacy et al. 2020) within 5″ ($3\sigma < 360\,\mu$Jy at 3 GHz; the source at Dec $= +3.5°$ is within the VLASS footprint) or in the Rapid ASKAP Continuum Survey (RACS; McConnell et al. 2020) within 10″ ($3\sigma < 750\,\mu$Jy at 887.5 MHz).

The source lies at Galactic longitude $l = 32.1°$, placing it in the eastern Galactic hemisphere assigned to the eROSITA-RU consortium. The publicly available eROSITA-DE eRASS1 catalogue (Merloni et al. 2024) covers only the western hemisphere ($l = 180$–$360°$), so no eROSITA data are available for this position.

2.4 SIMBAD classification

SIMBAD (Wenger et al. 2000) returns only a Tycho-2 match (TYC 456-894-1) classified as "Star" with no spec-

tral type, variability flag, or prior binarity designation. We also queried LAMOST DR9, GALAH DR3, RAVE DR6, and APOGEE DR17: none has observed this source, so no independent spectroscopic parameters are available from major stellar surveys. The primary's evolutionary state therefore rests entirely on Gaia GSP-Phot photometric parameters, which are known to be unreliable at high RUWE (Section 2.1).

3 METHODS

Standing caveat (referenced throughout). This source carries a purely astrometric `Orbital` solution with no independent radial-velocity orbit. The companion mass, all derived properties, and all scenario exclusions in this paper are *conditional on the Gaia pipeline model being a correct description of this system.* Population-level validation studies find that $\gtrsim$ 30–50 per cent of comparable `Orbital` solutions fail independent RV checks (El-Badry et al. 2024). Unless stated otherwise, every inference below should be read as prefixed by "if the orbital solution is genuine." Independent multi-epoch radial-velocity confirmation remains the single most important next step.

3.1 Gaia NSS mass derivation chain

The companion mass M_2 reported for this source originates from two distinct Gaia DR3 tables. The `gaiadr3.nss_two_body_orbit` table provides the astrometric orbital elements (period, eccentricity, angular semi-major axis, inclination, etc.) fitted to the along-scan residuals. The `gaiadr3.binary_masses` table then combines these elements with the parallax and an external primary-mass estimate to derive M_2 via Kepler's third law. The primary mass entering the `binary_masses` derivation is itself model-dependent: it relies on Gaia GSP-Phot atmospheric parameters and isochrone-based mass estimation.

Hence, while the `Orbital` solution eliminates the $\sin i$ degeneracy inherent to SB1 orbits — a critical advantage — the resulting M_2 is not a purely empirical "true mass" but depends on the modelling assumptions of the NSS pipeline, the parallax, and the adopted primary mass. We therefore quote the catalogue value $M_2 = 12.31\,\mathrm{M}_\odot$ (with formal uncertainty $\sigma_{M_2} = 7.52\,\mathrm{M}_\odot$) as a *derived mass under Gaia modelling assumptions* and propagate the relevant uncertainties independently via Monte Carlo (Section 3.6).

The large formal catalogue uncertainty ($\sigma_{M_2}/M_2 \approx$ 61 per cent) itself reflects the dominance of the parallax uncertainty when propagated correctly through Kepler's third law, as we demonstrate in Section 4.4.

El-Badry et al. (2024) have shown that a substantial fraction of extreme-mass-ratio candidates in the Gaia NSS catalogue are spurious when tested against independent radial velocities, particularly among `Orbital`-type solutions. The present source has *not* been flagged by any published vetting study, but the same failure modes could in principle apply. Independent spectroscopic confirmation therefore remains essential (Section 6.6).

3.2 Orbital dynamics

The total system mass follows from Kepler's third law:

$$M_{\rm tot} = M_1 + M_2 = \frac{4\pi^2\, a^3}{G\, P^2}\,, \tag{1}$$

where the true semi-major axis a derives from the angular semi-major axis and parallax. Because $a \propto \varpi^{-1}$, the total mass scales as $M_{\rm tot} \propto \varpi^{-3}\, P^{-2}$, making it extremely sensitive to parallax uncertainty: a 10 per cent parallax error maps to a $\sim$33 per cent mass error. This steep dependence is the dominant source of uncertainty in our posterior (Section 4.4).

3.3 Primary-mass estimation

The primary mass is estimated from the Gaia GSP-Phot parameters ($T_{\rm eff} = 5922\,\mathrm{K}$, $\log g = 2.93$). We note that GSP-Phot effective temperatures for RGB stars carry known systematic uncertainties of $\sim$100–200 K (Andrae et al. 2023), and that at RUWE = 9.31 the parameters may be additionally biased by unresolved orbital motion. No independent spectroscopic classification exists (Section 2.4), so the evolved-star identification rests entirely on pipeline photometric parameters. We adopt $M_1 = 1.34 \pm 0.40\,\mathrm{M}_\odot$, assigning a 30 per cent fractional uncertainty to encompass age, metallicity, and systematic errors; the sensitivity of our results to this choice is explored in Section 4.5.

As a cross-check, we construct a PARSEC-calibrated RGB mass grid (Bressan et al. 2012) spanning [Fe/H] $\in$ [−0.5, +0.3] and age $\tau \in$ [1, 12] Gyr at the catalogue $T_{\rm eff} =$ 5922 K and $\log g = 2.93$. The grid returns $M_1 \approx$ 0.76–2.26 $\mathrm{M}_\odot$, with higher masses at lower metallicity and younger age (script 13; Table A3). A Monte Carlo experiment drawing $T_{\rm eff}$, $\log g$, [Fe/H] and age from broad priors ($\sigma_{T_{\rm eff}} = 300\,\mathrm{K}$, $\sigma_{\log g} = 0.4$, uniform metallicity and flat age) yields a median $M_1 = 1.09\,\mathrm{M}_\odot$ with 68 per cent CI [0.77, 1.62] $\mathrm{M}_\odot$ and 90 per cent CI [0.63, 2.12] $\mathrm{M}_\odot$. The adopted $M_1 = 1.34 \pm 0.40\,\mathrm{M}_\odot$ falls within the central body of this distribution, confirming that it is a reasonable working estimate rather than an ad hoc choice. A full SED-based mass determination is not warranted given the RUWE-induced parameter uncertainty, but the independent route confirms that $M_1 = 1.34 \pm 0.40\,\mathrm{M}_\odot$ is a reasonable working estimate.

3.4 Extinction and SED self-consistency

The source lies at moderate Galactic latitude ($b = 9.3°$), where extinction is present but manageable. We derive the colour excess from the discrepancy between the observed $G_{\rm BP} - G_{\rm RP} = 0.993$ and the intrinsic colour for $T_{\rm eff} = 5922\,\mathrm{K}$ ($(G_{\rm BP} - G_{\rm RP})_0 \approx 0.76$):

$$E(G_{\rm BP} - G_{\rm RP}) = 0.993 - 0.76 = 0.233\,\mathrm{mag}\,, \tag{2}$$

yielding $E(B-V) = 0.175$ mag and $A_V = 0.543$ mag via the Cardelli et al. (1989) extinction law with $R_V = 3.1$.

We note a self-consistency tension in the primary's physical parameters. The *photometric* estimate (from the distance modulus, extinction-corrected bolometric magnitude, and a bolometric correction $BC_G \approx -0.3$ mag) yields $R_\star = 4.5\,\mathrm{R}_\odot$ and $L_\star = 22\,\mathrm{L}_\odot$. However, combining the adopted primary mass with the GSP-Phot surface gravity via $R = \sqrt{G\,M_1/g}$ gives a *logg-based* estimate of $R_\star = 6.5\,\mathrm{R}_\odot$ and $L_\star = 47\,\mathrm{L}_\odot$ — a factor $\sim$1.5× larger in radius and $\sim$2.1× in luminosity. This discrepancy may reflect the unreliability of $\log g$ at RUWE = 9.31 (Andrae et al. 2023), where unresolved orbital motion degrades the photometric solution. We adopt the photometric values ($R = 4.5\,\mathrm{R}_\odot$, $L = 22\,\mathrm{L}_\odot$) as our fiducial estimates but report both throughout and flag the discrepancy

as a systematic uncertainty to be resolved by independent spectroscopy.

3.5 SED fitting and synthetic photometry

We construct the spectral energy distribution from 8 photometric bands (Gaia G, $G_{\rm BP}$, $G_{\rm RP}$; 2MASS J, H, K_s; WISE $W1$, $W2$) after applying band-dependent extinction corrections. The dereddened SED is compared to a single-temperature model at $T_{\rm eff} = 5922\,$K. Residuals below 5 per cent in all bands show that the photometry is consistent with a single luminous star and no detectable secondary light contribution.

For the companion-exclusion analysis, we employ two independent approaches. First, Planck-based flux ratios are cross-checked against atmosphere-corrected values using multiplicative correction factors derived from ATLAS9 (Castelli & Kurucz 2003) and PHOENIX (Husser et al. 2013) temperature–bolometric-correction grids. These corrections are 2–13 per cent relative to the Planck ratio (Table A2).

Second, we perform a full synthetic-photometry analysis using parametrised atmosphere corrections derived from the same grids. For each band at effective wavelength $\lambda_{\rm eff}$, the correction factor is modelled as $f(T_{\rm eff}) = 1 + c_1\,(T_{\rm eff} - 5500)/(10^4) + c_2\,\log(\lambda/0.55\,\mu{\rm m})$, where c_1 and c_2 are calibrated against the published grids. The resulting *synthetic* companion-to-primary flux ratios for a $12.3\,{\rm M}_\odot$ MS companion are 100–247× in optical bands (photometric radius) — substantially larger than the Planck-only estimates — because the atmosphere models suppress the cool-star flux in the blue more strongly. The maximum main-sequence companion that remains hidden below a 5 per cent threshold tightens to $M \lesssim 0.70\,{\rm M}_\odot$ (photometric) / $0.85\,{\rm M}_\odot$ (logg-based), compared to $1.0/1.3\,{\rm M}_\odot$ from the Planck-only analysis.

We note a caveat on the extinction chain: our colour-excess derivation assumes the GSP-Phot $T_{\rm eff}$ is correct. If $T_{\rm eff}$ is biased by the high RUWE, then $E(B-V)$, the distance modulus, the luminosity, the radius, and the companion-exclusion limits are all affected downstream. This chain dependency cannot be broken without independent spectroscopic $T_{\rm eff}$.

To quantify the impact, we propagate $T_{\rm eff}$ offsets of ±250 and $\pm500\,$K through the full chain $T_{\rm eff} \to E(B-V) \to A_V \to L_\star \to R_\star \to$ max hidden MS mass (script 14). At $T_{\rm eff} = 5422\,$K ($-500\,$K), the primary radius swells to $R \approx 4.8\,{\rm R}_\odot$ and the maximum hidden MS companion rises to $\sim 0.96\,{\rm M}_\odot$; at $T_{\rm eff} = 6422\,$K ($+500\,$K), R shrinks to $\sim 3.5\,{\rm R}_\odot$ and the limit tightens to $\sim 0.88\,{\rm M}_\odot$. In all cases the MS companion at the catalogue mass ($12.3\,{\rm M}_\odot$) remains excluded by a factor $\gtrsim 5\times$ in every photometric band. The companion exclusion is therefore robust across the plausible range of GSP-Phot $T_{\rm eff}$ bias.

3.6 Monte Carlo mass posterior

We draw $N = 200\,000$ Monte Carlo realisations of the companion mass, propagating uncertainties in:

(i) **Parallax** ϖ: Gaussian ($1.5228 \pm 0.1549\,$mas), with an additional $\times1.7$ inflation factor applied to σ_ϖ to account for the known underestimation of external parallax errors for `Orbital`-type solutions at $G \approx 11$ (El-Badry et al. 2024).

(ii) **Period** P: Gaussian ($424.403 \pm 1.159\,$d).

(iii) **Primary mass** M_1: log-normal prior centred on $1.34\,{\rm M}_\odot$ with $\sigma = 0.40\,{\rm M}_\odot$.

(iv) **Model systematic**: uniform multiplicative factor [0.95, 1.05] on $M_{\rm tot}$.

The total mass scales as (Section 3.2):

$$M_{\rm tot} \propto \varpi^{-3}\,P^{-2}\,, \tag{3}$$

so the parallax uncertainty dominates. We emphasise that the $\times1.7$ inflation is an *approximate* correction; the true external-to-internal error ratio for this specific source is unknown. We adopt $\times1.7$ as the fiducial external-to-internal parallax inflation following published validation work for `Orbital` solutions in this magnitude regime (El-Badry et al. 2024), while Table A5 shows that our conclusions remain sensitive to the exact inflation factor assumed. We report posteriors both with and without inflation (Section 4.4).

Because M_2 is derived from the full astrometric orbit (not a minimum mass), no inclination marginalisation is needed. However, the resulting mass depends on all assumptions entering the Gaia NSS pipeline (Section 3.1). The Monte Carlo draws do not account for potential systematic errors in the orbital model itself (e.g. incorrect inclination from scan-geometry biases) — only for the quoted parameter uncertainties.

We emphasise that this posterior represents a *conditional working estimate* of the companion mass under the adopted assumptions, not a rigorously grounded Bayesian inference. In particular: (i) the $\times1.7$ inflation is calibrated at the population level, not for this individual source; (ii) the independent sampling of correlated parameters (necessitated by the unavailability of the full NSS covariance matrix) may underestimate the true uncertainty; and (iii) the uniform $[0.95, 1.05]$ multiplicative systematic is an ad hoc bracket whose range is not derived from first principles. A fully rigorous uncertainty model would require access to the epoch-level astrometric data and the full parameter covariance matrix, neither of which is publicly available in DR3.

3.7 Correlation sensitivity tests

Because Gaia DR3 does not publish the full covariance matrix for `Orbital` solutions, the Monte Carlo draws in Section 3.6 sample each parameter independently. To assess whether this simplification biases the posterior, we repeat the 2×10^5-draw experiment under several plausible inter-parameter correlation structures.

Parallax–angular-semi-major-axis correlation. For `Orbital` solutions the parallax and the angular semi-major axis of the photo-centre orbit share the same astrometric data stream, so a positive correlation is physically expected. We inject bivariate-normal correlations with $\rho = 0$, 0.3, 0.5, 0.7 between ϖ and the angular semi-major axis, holding all other parameters fixed. The resulting $P(M_2 > 5\,{\rm M}_\odot)$ *increases* monotonically: 91 per cent ($\rho = 0$), 94 per cent ($\rho = 0.3$), 97 per cent ($\rho = 0.5$), 99 per cent ($\rho = 0.7$). The direction of this effect is expected: both quantities share a common astrometric signal, so a positive correlation partially cancels when they enter the mass equation as a ratio, narrowing the posterior. Independent sampling is therefore *conservative* for this parameter pair.

Period–eccentricity correlation. We test $\rho = 0, 0.3, 0.5, 0.7$ between P and e. The impact is negligible ($\Delta P(M_2 > 5\,\mathrm{M}_\odot) < 0.1$ per cent), because the fractional period uncertainty ($\sigma_P/P \approx 0.3$ per cent) contributes negligibly to the mass variance relative to the parallax.

Systematic-bracket expansion. The uniform [0.95, 1.05] multiplicative systematic on $M_{\rm tot}$ was chosen as a conservative lower bound reflecting the typical level of pipeline-level mass-calibration residuals (Halbwachs et al. 2023). Expanding it to ±10 per cent and ±15 per cent changes $P(M_2 > 5\,\mathrm{M}_\odot)$ by only 0.2–0.4 percentage points, and the median M_2 is unchanged at $12.3\,\mathrm{M}_\odot$.

These tests confirm that the independent-sampling assumption is either conservative or has negligible impact, and that the posterior is robust to reasonable variations in the systematic bracket.

We note that negative ϖ–$a_{\rm ang}$ correlations are not physically expected for `Orbital` solutions (both parameters respond in the same direction to the astrometric signal), but if present they would *broaden* the posterior and lower $P(M_2 > 5\,\mathrm{M}_\odot)$. At $\rho = -0.3$ the BH probability drops by ~ 2 percentage points; the qualitative conclusion is unchanged, but we flag this as an additional source of systematic uncertainty that can only be resolved with the full NSS covariance matrix.

3.8 Companion-light exclusion

For a hypothetical main-sequence companion of mass M_c, we compute the expected luminosity using empirical mass–luminosity relations ($L \propto M^{3.5}$, $R \propto M^{0.57}$) and compare the resulting *band-by-band* flux ratio using Planck functions:

$$\frac{F_{\lambda,\,\rm comp}}{F_{\lambda,\,\rm prim}} = \left(\frac{R_{\rm comp}}{R_{\rm prim}}\right)^2 \frac{B_\lambda(T_{\rm comp})}{B_\lambda(T_{\rm prim})}\,. \tag{4}$$

As a practical heuristic, we consider a companion detectable if its flux exceeds 5 per cent of the primary in *any* photometric band. We perform this test using both the photometric ($R = 4.5\,\mathrm{R}_\odot$) and logg-based ($R = 6.5\,\mathrm{R}_\odot$) primary radii to bracket the self-consistency range.

4 RESULTS

4.1 Orbital and dynamical solution

Table A1 summarises the observed and derived properties. The orbital-solution significance (75.4σ) and astrometric excess-noise significance (3753σ) demonstrate that the single-star model is strongly rejected, but this does not by itself validate the Keplerian fit: similar significance levels can arise from poorly resolved blends or pipeline failures. The coherent Keplerian fit and well-constrained period precision (0.27 per cent) do not exclude such explanations. *Independent RV confirmation is therefore essential.*

The semi-major axis $a = 2.64\,\mathrm{AU}$ places the companion well outside the primary's Roche lobe ($R_L \approx 0.56\,\mathrm{AU}$; filling factor ∼3.7 per cent), consistent with a fully detached, non-interacting system. The eccentricity $e = 0.34$ is tidally consistent: the circularisation timescale $\tau_{\rm circ} \gg \tau_{\rm Hubble}$, so the orbit retains its natal eccentricity.

The predicted RV semi-amplitude is $K_1 \approx 65\,\mathrm{km\,s^{-1}}$, readily detectable with ground-based spectrographs (Section 6.6).

4.2 Stellar characterisation

The extinction analysis reveals modest reddening ($A_V = 0.54\,\mathrm{mag}$). The dereddened absolute magnitude $M_G = 1.68\,\mathrm{mag}$ and the photometric luminosity $L_\star = 22\,\mathrm{L}_\odot$ are consistent with the GSP-Phot evolved-star classification adopted as a working assumption, with radius $R_\star = 4.5\,\mathrm{R}_\odot$. The alternative logg-based estimates ($R = 6.5\,\mathrm{R}_\odot$, $L = 47\,\mathrm{L}_\odot$) differ by factors of $\sim 1.5\times$ (radius) and $\sim 2.1\times$ (luminosity). This is not a cosmetic discrepancy: it indicates that the stellar characterisation feeding the companion interpretation is *provisional*. The tension may reflect unreliable GSP-Phot parameters at the source's high RUWE (Section 3.4), and demonstrates that neither estimate should be treated as a secure physical measurement absent independent spectroscopy. The dereddened SED is well-fit by a single-temperature model with residuals < 5 per cent in all bands. No secondary SED component is required.

4.3 Companion-light exclusion

A main-sequence star of $M_2 = 12.3\,\mathrm{M}_\odot$ would have $L \approx 6550\,\mathrm{L}_\odot$, $R \approx 4.2\,\mathrm{R}_\odot$, and $T_{\rm eff} \approx 25\,400\,\mathrm{K}$. Using atmosphere-corrected band-by-band flux ratios (Section 3.5) with the photometric primary radius ($R = 4.5\,\mathrm{R}_\odot$), the companion-to-primary flux ratio would be $\sim 55\times$ in $G_{\rm BP}$, $\sim 29\times$ in G, and $\sim 18\times$ in $G_{\rm RP}$ (Table A2; under the full synthetic-photometry model these ratios rise to 100–247× in optical bands). Even adopting the larger logg-based radius ($R = 6.5\,\mathrm{R}_\odot$), the ratios remain $\sim 26\times$ in $G_{\rm BP}$ and $\sim 14\times$ in G — overwhelmingly detectable.

The maximum main-sequence companion that could remain hidden below a 5 per cent detection threshold in all bands is $M \lesssim 1.0\,\mathrm{M}_\odot$ (photometric case) to $\sim 1.3\,\mathrm{M}_\odot$ (logg-based case). A normal main-sequence companion at the catalogue mass is therefore firmly excluded under the adopted solution.

4.4 Mass posterior

As noted in the standing caveat (Section 3), the posteriors below are conditional working estimates that do not account for systematic errors in the `Orbital` model itself.

The Monte Carlo posterior (Fig. A2) yields:

Nominal (σ_ϖ as published):

- Median $M_2 = 12.3\,\mathrm{M}_\odot$
- 68 per cent CI: $[8.8, 17.5]\,\mathrm{M}_\odot$
- 90 per cent CI: $[7.1, 22.4]\,\mathrm{M}_\odot$
- $P(M_2 > 5\,\mathrm{M}_\odot) = 99.7$ per cent; $P(M_2 > 10\,\mathrm{M}_\odot) = 73$ per cent

Inflated ($\sigma_\varpi \times 1.7$):

- Median $M_2 = 12.3\,\mathrm{M}_\odot$
- 68 per cent CI: $[7.1, 22.7]\,\mathrm{M}_\odot$
- 90 per cent CI: $[5.0, 35.9]\,\mathrm{M}_\odot$
- $P(M_2 > 5\,\mathrm{M}_\odot) = 95$ per cent; $P(M_2 > 10\,\mathrm{M}_\odot) = 64$ per cent

The wide credible intervals reflect the ϖ^{-3} mass scaling (equation 3): a 10 per cent parallax error maps to a

$\sim$33 per cent mass error, and the $\times$1.7 inflation broadens this substantially. It is important to note that the nominal posterior already has a lower 5th-percentile above 5 $M_\odot$; with inflation, $\sim$5 per cent of draws fall below the canonical BH threshold.

More fundamentally, the Monte Carlo posterior is best understood as a mixture of two components: (i) a "genuine-solution" component, under which the smooth posterior around $M_2 \sim 12\,M_\odot$ is meaningful, and (ii) a "spurious" component (probability $\sim$ 30–50 per cent; Section 6.2), under which the orbital parameters are not physically reliable and the companion mass is effectively unconstrained (i.e. the companion could range from non-existent to any mass). Marginalising over both components, the credible intervals reported above describe only the first mode. Readers should therefore interpret the CIs as conditional on the solution being genuine; the overall probability assigned to the BH hypothesis must also account for the non-trivial probability that the orbital solution is spurious. This two-component framing replaces any single-number "effective BH probability" and makes explicit the dominant source of uncertainty.

A parallax stress test illustrates the sensitivity directly: at $\varpi + 1\sigma$ (1.678 mas, $d = 596$ pc), the companion mass drops to $M_2 \approx 8.9\,M_\odot$; at $\varpi - 1\sigma$ (1.368 mas, $d = 731$ pc), it rises to $M_2 \approx 17.5\,M_\odot$.

4.5 Sensitivity to primary mass

Table A5 shows how M_2 and the BH probability vary across a grid of M_1 values (0.8–2.0 $M_\odot$) and parallax inflation factors (1.0–2.0$\times$). At the fiducial values ($M_1 = 1.34\,M_\odot$, inflation = 1.7), $P(M_2 > 5\,M_\odot) = 95$ per cent. For inflation factors ≥ 2.0, the BH probability drops below 90 per cent, highlighting the importance of improved parallax precision from Gaia DR4. Fig. A3 visualises the full sensitivity landscape.

5 ALTERNATIVE SCENARIOS

We systematically evaluate seven non-BH explanations for the dark companion. All verdicts are subject to the standing caveat (Section 3): if the orbital solution is spurious, these exclusions lose their force. Table A4 summarises the results, using a four-tier verdict system: **Excluded** (physically impossible or overwhelmingly contradicted by data if the orbital solution is genuine), **Strongly disfavoured** (requires implausible fine-tuning under the adopted assumptions), **Not favoured** (unlikely but not formally excluded), and **Requires follow-up** (cannot be assessed without additional data).

Main-sequence companion. A 12.3 $M_\odot$ MS star would have $L \sim 6550\,L_\odot$ and $T_{\rm eff} \sim 25\,400$ K, contributing $\sim$29$\times$ the primary flux in G and $\sim$55$\times$ in $G_{\rm BP}$ (atmosphere-corrected; 100–247$\times$ under synthetic photometry; Table A2). No secondary SED component is detected. **Excluded under adopted solution.**

White dwarf. $M_2 = 12.3\,M_\odot$ exceeds the Chandrasekhar limit (1.44 $M_\odot$) by a factor of $\sim$9. **Excluded under adopted solution.**

Neutron star. $M_2 = 12.3\,M_\odot$ exceeds the most generous NS mass ceiling ($\sim$2.3 $M_\odot$; Rezzolla et al. 2018) by a factor of $\sim$5. **Excluded under adopted solution.**

Hierarchical triple. Two $\sim$6.2 $M_\odot$ MS stars would have combined luminosity $\sim$1160 $L_\odot$ — easily detectable in the SED. The Mardling–Aarseth stability criterion (Mardling & Aarseth 2001) would require $P_{\rm inner} \lesssim 42$ d; no evidence for such a configuration is currently available from the existing photometric and astrometric data, though dedicated spectroscopic and time-domain observations have not yet been obtained. **Strongly disfavoured.**

Stripped helium star / hot subdwarf. At the median posterior mass of 12.3 $M_\odot$, a stripped He star would have $T_{\rm eff} > 50\,000$ K and $L \sim 10^4\,L_\odot$ (Götberg et al. 2018), producing predicted NUV $\sim$ 9 mag vs. observed NUV = 17.52 — a flux deficit of $\sim$2700$\times$ and an optical excess of $\sim$38$\times$. To test whether a *lower-mass* stripped core could hide in the photometry, we scan the full range 1–15 $M_\odot$ using the mass–luminosity and mass–radius relations of Götberg et al. (2018). No stripped He star in this range is photometrically compatible with the observed SED: even a 1 $M_\odot$ stripped core ($T_{\rm eff} \approx 42\,000$ K, $R \approx 0.4\,R_\odot$) would produce an optical excess of $\sim$80 per cent of the primary flux, well above the 5 per cent detectability threshold. No X-ray source is detected. This scenario is **not favoured** across the entire plausible mass range, but a sub-solar-mass hot subdwarf falling outside the Götberg et al. (2018) calibration cannot be fully excluded without UV spectroscopy. We note that the Götberg et al. (2018) mass–luminosity relations are calibrated at solar metallicity; at sub- or super-solar metallicity the envelope-stripping efficiency and residual hydrogen content shift modestly, potentially altering the predicted optical flux by tens of per cent without qualitatively changing the exclusion. Specifically, at [Fe/H] = -0.5 the opacity-driven change in envelope-stripping efficiency shifts $\log(L/L_\odot)$ by $\sim +0.1$–0.2 dex and $T_{\rm eff}$ by $\sim +2000$–3000 K relative to the solar calibration (Götberg et al. 2018), increasing the predicted optical flux ratio by a factor of $\sim$ 1.5–2 — i.e. the stripped-star scenario becomes *easier* to exclude at sub-solar metallicity. At super-solar metallicity ([Fe/H] $\approx +0.3$) the effect reverses but more modestly ($\sim$ 20–30 per cent lower flux ratio), still leaving the stripped star well above the 5 per cent detectability threshold for masses $\gtrsim 1\,M_\odot$.

Astrometric artefact (spurious orbital solution). Because the solution type is `Orbital` (astrometry only), there is no independent spectroscopic confirmation of the orbital parameters. El-Badry et al. (2024) have demonstrated that a fraction of high-significance `Orbital` solutions fail independent RV tests. While the 75.4σ significance, the well-constrained period, and the low GOF (18.3) do not by themselves rescue the solution from this concern: this scenario **cannot be excluded** without independent RV data. This is the single most important limitation of the present analysis.

Chance alignment / blend-induced artefact. At G = 11.25 and $b = 9.3°$, source confusion is low but not negligible. A coherent 424-d Keplerian signal at 75σ is difficult to produce from a chance blend, but scan-angle correlations

with a nearby source could in principle mimic orbital motion. This scenario likewise **requires follow-up** via independent RV monitoring and high-resolution imaging.

Summary (subject to standing caveat, Section 3): three of seven alternative scenarios are photometrically inconsistent (MS companion, WD, NS); one (hierarchical triple) is strongly disfavoured; one (stripped He star) is not favoured but remains formally open. Two scenarios — astrometric artefact and chance alignment — *cannot be excluded* at present. A BH companion is *not excluded* by the archival data, but this reflects the absence of contradicting evidence rather than positive confirmation.

6 DISCUSSION

6.1 Evidence hierarchy

To clarify the epistemic status of each element of our analysis, we distinguish four tiers of evidence:

(i) **Directly measured.** Observed multi-band photometry (G, $G_{\rm BP}$, $G_{\rm RP}$, 2MASS, WISE, GALEX NUV), parallax ($\varpi = 1.5228 \pm 0.1549$ mas), systemic radial velocity ($v_{\rm sys} = -29.49 \pm 0.57\,{\rm km\,s^{-1}}$), and ROSAT/VLASS/RACS non-detections. These are empirical observables, subject only to standard calibration uncertainties.

(ii) **Pipeline-derived (model-dependent).** The Gaia NSS orbital parameters (period, eccentricity, angular semi-major axis, inclination) from the `Orbital` pipeline fit to along-scan astrometric residuals. The GSP-Phot stellar parameters ($T_{\rm eff}$, $\log g$), which are known to be unreliable at RUWE = 9.31 (Andrae et al. 2023). These quantities depend on the Gaia data reduction model and may not reflect the true system properties.

(iii) **Assumed.** The primary mass ($M_1 = 1.34 \pm 0.40\,{\rm M}_\odot$) from isochrone matching of GSP-Phot parameters. The $\times 1.7$ parallax-error inflation factor, adopted from population-level validation studies (El-Badry et al. 2024) but not verified for this individual source. The uniform $[0.95, 1.05]$ multiplicative systematic on total mass. The single-temperature SED model (cross-checked against atmosphere corrections; Section 3.5).

(iv) **Conditional inference.** The companion mass posterior ($M_2 = 12.3\,{\rm M}_\odot$), the dark-companion classification, and all alternative-scenario exclusions. These are valid *only if* the Gaia `Orbital` solution correctly describes the system and the adopted stellar model is approximately correct. If the orbital solution is spurious (Section 6.2) or the primary mass is substantially different from the adopted value, the entire downstream inference changes accordingly.

This hierarchy makes explicit that the paper's central result — the inferred high companion mass — sits at the fourth tier (see standing caveat, Section 3).

6.2 Reliability of the Gaia orbital solution

The reliability of the `Orbital` solution is the most consequential uncertainty in this analysis. We assess it against several available diagnostics, while emphasising that none of these constitutes a substitute for independent RV confirmation.

Solution quality metrics. The solution significance (75.4σ) and astrometric excess-noise significance (3753σ) indicate that the single-star model is strongly rejected. This is a necessary but not sufficient condition for a genuine orbital signal: similar significance levels arise in confirmed binaries but also in sources with poorly modelled systematics. The goodness-of-fit (GOF = 18.3) exceeds the ideal value of ~ 0, indicating that the single-Keplerian model leaves residual scatter; for context, Halbwachs et al. (2023) find GOF values of 5–30 for validated binaries at comparable RUWE, so the present value is not anomalous but also does not provide strong discriminative power between genuine and spurious solutions.

Fraction of spurious `Orbital` solutions. El-Badry et al. (2024) tested a sample of high-mass-ratio `Orbital` candidates against independent RV data and found that a substantial fraction ($\gtrsim$ 30–50 per cent, depending on selection criteria) showed RV semi-amplitudes inconsistent with the astrometric prediction. No systematic vetting of this specific source has been published. Applying the population-level false-positive rate as an informal prior suggests a non-trivial probability that the orbital solution does not reflect the true system dynamics. This base-rate concern is the principal motivation for framing this source as a candidate requiring confirmation rather than a secure detection.

Scan-angle coverage. For `Orbital` solutions with periods near one year, the Gaia scanning law can produce near-degenerate scan-angle distributions that permit spurious orbital fits. The present system has $P = 424.4$ d (~ 1.16 yr), within the ~ 0.8–1.5 yr window where annual scan-angle aliasing is a documented concern. Without access to the epoch-level astrometric data (not publicly available in DR3), we cannot quantify the scan-angle distribution or assess whether the orbital fit is driven by a limited range of position angles. This remains an unresolved systematic that can only be addressed with the epoch-level data expected in Gaia DR4.

Gaia DR4 is expected to release per-transit astrometric measurements and the full NSS covariance matrices currently withheld. These data would enable three critical diagnostics: (i) direct visualisation of the scan-angle distribution, revealing whether the orbital signal arises from a well-sampled range of position angles or is driven by a few clustered transits; (ii) epoch-level residual analysis, testing whether the orbital fit is dominated by outliers; and (iii) access to the ϖ–$a_{\rm ang}$ correlation coefficient, removing the need for the assumed correlation structures explored in Section 3.7. Together, these would either substantially strengthen the candidacy or reveal the solution as an artefact of the scanning geometry.

Blend probability. At $G = 11.25$ and Galactic latitude $b = 9.3°$, the local stellar density is moderate. The Gaia archive reports no resolved companion brighter than $G \sim 20$ within $2''$. However, a faint unresolved blend at sub-arcsecond separations could influence the astrometric solution. High-resolution imaging (Section 6.6) is needed to assess this possibility quantitatively.

Assessment. The available diagnostics do not validate the solution and do not rule it out. Given that El-Badry et al.

(2024) find $\gtrsim$ 30–50 per cent of `Orbital` high-mass-ratio solutions fail independent RV tests, the prior probability that any given unvalidated `Orbital` candidate is genuine is at most $\sim$50–70 per cent before source-specific information is considered. For this source, the near-annual period (1.16 yr) adds further caution. We therefore regard the orbital solution as not excluded by available diagnostics but unverified. This assessment underlies our framing of the entire analysis as conditional and our emphasis on spectroscopic confirmation as the essential next step.

Effective BH probability. Folding the false-positive rate into the mass posterior gives an order-of-magnitude "effective" BH probability:

$$P_{\rm eff}({\rm BH}) \approx P({\rm genuine}) \times P(M_2 > 5\,{\rm M}_\odot \mid {\rm genuine}) \approx 0.5\text{–}0.7 \times 0.95 \approx 48\text{–}67 \text{ per cent}\,. \quad (5)$$

This should be interpreted as an informal Bayesian estimate, not a rigorous posterior: P(genuine) $\approx$ 0.5–0.7 is the complement of the population-level false-positive rate and is not calibrated for this individual source. The estimate makes explicit that, even under the most favourable reading of the archival data, the probability that this source genuinely hosts a BH companion is of order one-in-two — underscoring the necessity of RV confirmation.

6.3 Why this target

The Gaia DR3 NSS catalogue contains hundreds of `Orbital` solutions with formally high companion masses, the majority of which are expected to be spurious (El-Badry et al. 2024). To justify highlighting a single source for follow-up, we quantitatively compare it against the broader `Orbital` BH-candidate population.

The 47-source working catalogue is constructed by querying `gaiadr3.nss_two_body_orbit` for all entries with `nss_solution_type = 'Orbital'`, cross-matching with `gaiadr3.binary_masses` on `source_id`, and retaining sources whose pipeline-reported companion mass satisfies $M_2 > 5\,{\rm M}_\odot$. Quality tiers (Gold/Silver/Bronze) are assigned following the vetting flags of Gaia Collaboration et al. (2023b): Gold requires GOF < 25, significance $> 20\sigma$, and eccentricity error $\sigma_e < 0.1$; Silver relaxes one of these; Bronze relaxes two or more. The full machine-readable table is provided in the repository (Section 7).

For each of these 47 sources we compute the predicted *edge-on* RV semi-amplitude

$$K_{1,\rm edge} = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{M_2}{(M_1+M_2)^{2/3}} \frac{1}{\sqrt{1-e^2}}\,, \quad (6)$$

which assumes $\sin i = 1$ and serves as a standardised ranking metric independent of inclination. For this source, the Gaia-derived orbital inclination implies $\sin i$ close to unity, so the source-specific predicted semi-amplitude $K_1 \approx K_{1,\rm edge} \approx 65\,{\rm km\,s^{-1}}$; we use K_1 (without the "edge" subscript) elsewhere in this paper to refer to this source-specific value.

Table A6 ranks the 16 brightest ($G < 12$) among these 47 candidates by $K_{1,\rm edge}$.

This source ranks **1st among 16 bright ($G < 12$) Orbital BH candidates** and 7th overall (out of 47) by $K_{1,\rm edge}$. The candidates with higher $K_{1,\rm edge}$ are all ≥ 1.2 mag fainter, requiring significantly longer integration times for the same RV precision. This source also has the highest NSS solution significance (75.4σ) among the top ten ranked candidates, and its equatorial declination (Dec $= +3.5°$) makes it accessible from both hemispheres. None of the 16 bright candidates in Table A6 has been subjected to published RV vetting, so the relative ranking is among unvalidated sources.

We emphasise that *none* of these ranking criteria validate the orbital solution. They quantify observational accessibility and predicted signal strength under the adopted model; the catalogue itself is unvetted, and individual rankings could change substantially if, for example, the orbital solution for any source is subsequently found to be spurious. The ranking should be treated as a follow-up *prioritization metric*, not as evidence for a genuine BH.

6.4 Context among Gaia BH discoveries

Table A7 lists the orbital and stellar parameters of this system alongside the three confirmed Gaia dormant BH systems for reference. We include this comparison solely to provide context for the predicted parameter space; it should not be read as implying equivalent evidentiary standing.

There is a *fundamental evidentiary gap* between this candidate and the three confirmed systems. Gaia BH1, BH2, and BH3 each benefited from independent multi-epoch radial-velocity follow-up that validated the astrometric orbital parameters and confirmed the companion mass. In contrast, the present system has only a purely astrometric `Orbital` solution with no RV confirmation. All entries in the "This work" column of Table A7 are therefore *conditional* on the Gaia pipeline assumptions and must not be compared on equal footing with the dynamically confirmed values for BH1–3.

If the orbital solution is confirmed, the long period ($P = 424$ d) and moderate eccentricity ($e = 0.34$) would place it in a similar orbital regime to Gaia BH1 ($P = 185.6$ d, $e = 0.45$) and Gaia BH2 ($P = 1277$ d, $e = 0.52$).

6.5 Binary evolution implications (contingent on confirmation)

If the companion is indeed a $\sim 12\,{\rm M}_\odot$ BH, the progenitor was a massive star with zero-age main-sequence mass $M_{\rm ZAMS} \gtrsim 25\,{\rm M}_\odot$ (Fryer et al. 2012). The wide orbit ($a = 2.64$ AU) would have survived the supernova event, constraining the natal kick to moderate values. No spectroscopic abundance analysis is yet available. Independent high-resolution spectroscopy will be required to test for any abundance anomalies associated with prior mass transfer; if confirmed, the absence of significant enrichment would suggest minimal pre-collapse mass transfer, favouring a direct-collapse or failed-supernova channel (Langer et al. 2020). These inferences are subject to the standing caveat (Section 3).

6.6 Follow-up strategy

The predicted RV semi-amplitude $K_1 \approx 65\,{\rm km\,s^{-1}}$ is large, making spectroscopic confirmation feasible with modest ground-based resources. We outline the critical follow-up steps in order of priority:

(i) **Multi-epoch radial-velocity monitoring** (highest priority). At $G = 11.25$, the source is within reach of echelle spectrographs on 2–4 m class telescopes (e.g. HARPS, FEROS, CHIRON). We recommend a minimum of **8–10 RV epochs** distributed across one full orbital period (~ 424 d), with phase coverage concentrated near the predicted periastron passage (where $\dot{v}_r$ is maximal). At the predicted $K_1 \approx 65\,\mathrm{km\,s^{-1}}$, even a precision of ~ 0.5–$1\,\mathrm{km\,s^{-1}}$ per epoch suffices to confirm the period, eccentricity, and semi-amplitude. Even a single well-timed epoch could reveal significant tension with the Gaia systemic velocity and motivate further monitoring, though establishing the full RV curve requires the recommended multi-epoch campaign.

The RV orbit independently yields $M_2 \sin i$ via the spectroscopic mass function; combined with the astrometric inclination, this provides a direct, pipeline-independent companion mass.

(ii) **Spectroscopic classification.** High-resolution spectroscopy would yield independent $T_{\rm eff}$, $\log g$, metallicity, detailed abundances (testing for SN-related CNO anomalies), and a spectral type — resolving the current reliance on GSP-Phot parameters.

(iii) **UV spectroscopy.** HST/COS or Swift/UVOT observations would definitively test the stripped He star scenario by searching for hot-companion UV excess.

(iv) **High-resolution imaging.** AO-assisted imaging could detect or rule out a blended optical companion responsible for a scan-angle-correlated astrometric artefact.

The target is located at RA $= 18^{\rm h}28^{\rm m}$, Dec $= +3.5°$ (Ophiuchus), accessible from both hemispheres. At the stated right ascension the source transits in the evening during northern-hemisphere summer (approximately May–September) and is best observed from southern sites during their winter months.

6.7 Limitations

We stress the following limitations, which collectively prevent this system from being classified as a confirmed BH:

(i) **`Orbital`-only solution.** The companion mass depends entirely on the Gaia astrometric pipeline model. No independent RV orbit exists. This is the single most important limitation and distinguishes this source from confirmed Gaia BH systems.

(ii) **Parallax-driven uncertainty.** With $\sigma_\varpi/\varpi = 10.2$ per cent and $M_{\rm tot} \propto \varpi^{-3}$, the mass uncertainty is substantial. The external parallax-error inflation ($\times 1.7$) is an approximate correction; the true systematic for this source is unknown.

(iii) **NSS covariance matrix unavailable.** Gaia DR3 does not publish the full covariance matrix of the orbital parameters. Our Monte Carlo independently samples each parameter; however, correlation sensitivity tests (Section 3.7) show that plausible positive correlations *narrow* the posterior, so the independent-sampling assumption is conservative.

(iv) **GSP-Phot parameters unreliable at high RUWE.** At RUWE $= 9.31$, the primary's $T_{\rm eff}$ and $\log g$ may be biased. The R/L self-consistency tension (Section 3.4) exemplifies this.

(v) **Primary mass provisional.** No independent spectroscopic mass determination exists. The adopted $M_1 = 1.34 \pm 0.40\,\mathrm{M_\odot}$ is isochrone-derived from potentially biased GSP-Phot parameters.

(vi) **Scan-angle bias possible.** For `Orbital` solutions, scan-angle correlations with nearby sources or detector-level systematics could produce spurious orbital signals. This cannot be assessed without the epoch-level astrometric data (not publicly available in DR3).

(vii) **Stripped He star not formally excluded.** The GALEX NUV non-excess and X-ray non-detections strongly disfavour this scenario but cannot definitively rule it out without UV spectroscopy.

(viii) **Approximate SED model.** Our primary SED check uses a single-temperature model with atmosphere corrections; a supplementary synthetic-photometry analysis using ATLAS9/PHOENIX corrections (Section 3.5) strengthens the companion exclusion but still relies on parametrised rather than full spectral fitting. A full synthetic-atmosphere fit would be preferable for characterising the primary and detecting faint secondary contributions.

7 CONCLUSIONS

We have presented an archival multi-wavelength assessment and follow-up prioritization of Gaia DR3 4277855016732107520, identifying it as a high-priority spectroscopic follow-up target from the Gaia DR3 NSS catalogue. The source carries a purely astrometric `Orbital` solution with no independent RV orbit; all inferred companion properties are conditional on the Gaia pipeline modelling assumptions, and the population-level false-positive rate for such solutions is $\gtrsim$ 30–50 per cent (El-Badry et al. 2024). Our principal findings are:

(i) **Conditional companion mass.** Under the Gaia NSS modelling assumptions and adopted stellar model, the companion mass posterior yields a median $M_2 = 12.3\,\mathrm{M_\odot}$ with 68 per cent CI $[7.1, 22.7]\,\mathrm{M_\odot}$ (with $\times 1.7$ parallax-error inflation). Correlation sensitivity tests (Section 3.7) show that the independent-parameter sampling is conservative: introducing plausible positive ϖ–$a_{\rm ang}$ correlations *narrows* the posterior. The posterior should not be interpreted as a rigorous Bayesian inference.

(ii) **Companion-light exclusion.** Synthetic-photometry flux ratios using ATLAS9/PHOENIX atmosphere models show that a main-sequence star at the inferred mass would exceed the primary flux by 100–247$\times$ in optical bands. The maximum hidden MS companion tightens to $\lesssim 0.70\,\mathrm{M_\odot}$ (photometric radius), and no stripped helium star from 1 to $15\,\mathrm{M_\odot}$ is photometrically compatible with the observed SED.

(iii) **Alternative scenarios (conditional).** Of seven non-BH scenarios, three are photometrically inconsistent with the adopted solution (MS companion, WD, NS), one is strongly disfavoured (triple), and one is not favoured (stripped He star). Two scenarios — astrometric artefact and chance alignment — *cannot be excluded* and represent the dominant source of systematic uncertainty.

(iv) **Multi-wavelength non-detections.** ROSAT non-detection implies $\log L_X \lesssim 30.7\,\mathrm{erg\,s^{-1}}$; VLASS and RACS show no radio counterpart. These are consistent with a quiescent system but do not constitute positive evidence for a BH.

(v) **Stellar characterisation provisional.** The primary's physical parameters rest on GSP-Phot values obtained at RUWE = 9.31, a regime where systematic biases are known. The factor-of-two luminosity tension between photometric and logg-based estimates (Section 3.4) underscores that the stellar characterisation is provisional.

(vi) **Follow-up feasibility and population ranking.** The predicted $K_1 \approx 65\,\mathrm{km\,s^{-1}}$ makes multi-epoch RV confirmation feasible with 2–4 m class facilities over $\sim$8–10 epochs spanning one orbital period (Section 6.3). Among the 16 bright ($G < 12$) `Orbital` BH candidates in our working catalogue, this source ranks first by predicted K_1 (Table A6). The ranking reflects observational accessibility within an unvetted catalogue, not evidence for the reality of any solution.

(vii) **Overall assessment.** As a rough guide, the population-level false-positive rate ($\sim$ 30–50 per cent; Section 6.2) means that even the most favourable reading of the archival data leaves the probability of a genuine BH companion at approximately even odds (Equation 5). This order-of-magnitude estimate should not be interpreted as a calibrated posterior; it folds an informal population prior into a conditional mass posterior and serves only to underscore the necessity of RV confirmation.

We conclude that Gaia DR3 4277855016732107520 is a high-priority spectroscopic follow-up target: the archival data do not rule out a massive dark companion, and the observational properties (K_1, brightness, accessibility) make confirmation or refutation efficient with existing ground-based facilities. However, the orbital solution is unverified, the stellar characterisation is provisional, and the statistical framework is approximate. Until independent RV monitoring confirms or refutes the orbital parameters, the companion mass and nature remain entirely provisional, and no astrophysical conclusions should be drawn from the conditional mass estimate alone.

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DATA AVAILABILITY

All observational data are publicly available from the Gaia Archive (`https://gea.esac.esa.int/archive/`), VizieR, and SIMBAD. The analysis scripts and machine-readable results are available at `https://github.com/jayalabaez/dr3-4277855-black-hole-candidate`. Modular analysis scripts (`scripts/02–15`) produce all figures and numerical results, including the correlation sensitivity tests (script 11), synthetic-photometry SED analysis (script 12), the PARSEC isochrone grid (script 13), the $T_{\rm eff}$ chain-dependency sensitivity (script 14), the source-ID recovery cross-match (script 15), and the population-level comparison (script 10; Table A6). The repository includes the exact scripts used to generate all figures, tables, and machine-readable posterior summaries from the public catalogue inputs. All numerical results are in JSON format under `results/`.

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APPENDIX A: NOTE ON SCOPE AND CLAIMS

This paper is a single-object follow-up dossier and observing-proposal guide. It does *not* claim the discovery of a black hole. The companion mass, the dark-companion classification, and all scenario exclusions presented herein are conditional on the unverified Gaia `Orbital` solution and on the adopted primary-star model (standing caveat, Section 3). Until independent multi-epoch radial-velocity monitoring confirms or refutes the orbital parameters, no astrophysical conclusion

about the companion nature should be drawn from the conditional mass estimate alone.

Table A1. Observed and derived properties of Gaia DR3 4277855016732107520. Values marked with † depend on Gaia NSS modelling assumptions (Section 3.1).

Property	Value	Uncertainty	Source
	Astrometry (Gaia DR3)		
RA (J2016.0)	274.2238°	—	Gaia DR3
Dec (J2016.0)	+3.4975°	—	Gaia DR3
l, b	32.12°, +9.28°	—	Gaia DR3
Parallax ϖ	1.5228 mas	0.1549	Gaia DR3
Distance	$\sim$657 pc	$\sim$67	$1/\varpi$
G	11.25 mag	0.003	Gaia DR3
$G_{\rm BP}$	11.66 mag	0.003	Gaia DR3
$G_{\rm RP}$	10.66 mag	0.004	Gaia DR3
$G_{\rm BP} - G_{\rm RP}$	0.993 mag	—	Gaia DR3
RUWE	9.31	—	Gaia DR3
Astr. excess noise sig.	3753σ	—	Gaia DR3
	NSS orbital solution (`Orbital`)		
Period P	424.403 d	1.159	Gaia NSS
Eccentricity e	0.3427	0.0194	Gaia NSS
NSS significance	75.4σ	—	Gaia NSS
GOF	18.3	—	Gaia NSS
M_2 (catalogue)†	12.31 $M_\odot$	7.52	Gaia `binary_masses`
	Monte Carlo posterior (this work)		
M_2 (nominal)†	12.3 $M_\odot$	68% CI: [8.8, 17.5]; 90% CI: [7.1, 22.4]	This work
M_2 (inflated $\times 1.7$)†	12.3 $M_\odot$	68% CI: [7.1, 22.7]; 90% CI: [5.0, 35.9]	This work
$P(M_2 > 5\,M_\odot)$ (inflated)	95%	—	This work
$P(M_2 > 10\,M_\odot)$ (inflated)	64%	—	This work
	Stellar parameters (GSP-Phot; caution: RUWE = 9.31)		
$T_{\rm eff}$	5922 K	$\sim$200	Gaia DR3
$\log g$	2.93 dex	$\sim$0.3	Gaia DR3
$v_{\rm sys}$	-29.49 km s^{-1}	0.57	Gaia DR3
	Multi-wavelength photometry		
J (2MASS)	9.84 mag	0.02	2MASS
H (2MASS)	9.42 mag	0.02	2MASS
K_s (2MASS)	9.33 mag	0.02	2MASS
$W1$ (WISE)	9.23 mag	0.02	AllWISE
$W2$ (WISE)	9.27 mag	0.02	AllWISE
NUV (GALEX)	17.52 mag	—	GALEX
	Derived quantities		
$E(B-V)$	0.175 mag	0.03	Colour excess
A_V	0.543 mag	0.09	$R_V = 3.1$
M_G (dereddened)	1.68 mag	0.3	Derived
$R_\star$ (phot.)	4.5 $R_\odot$	—	Photometric
$L_\star$ (phot.)	22 $L_\odot$	—	Photometric
$R_\star$ ($\log g$)	6.5 $R_\odot$	—	logg-based
$L_\star$ ($\log g$)	47 $L_\odot$	—	logg-based
M_1 (adopted)	1.34 $M_\odot$	0.40	GSP-Phot + isochrone
a (semi-major axis)	2.64 AU	—	Kepler III
K_1 (predicted)	$\sim$65 km s^{-1}	—	Dynamics

Table A2. Companion-to-primary flux ratios for a 12.3 $M_\odot$ MS companion ($T_{\rm eff}$ = 25 400 K, R = 4.2 $R_\odot$) relative to the primary ($T_{\rm eff}$ = 5922 K), computed using both photometric and logg-based primary radii. Columns labelled "Planck+atm" use Planck functions with ATLAS9/PHOENIX atmosphere corrections (2–13 per cent; bare Planck values in parentheses). Columns labelled "Synth." use the full parametrised synthetic-photometry model of Section 3.5.

Band	Planck+atm		Synthetic photometry	
	Phot. ($R = 4.5\,R_\odot$)	$\log g$ ($R = 6.5\,R_\odot$)	Phot. ($R = 4.5\,R_\odot$)	$\log g$ ($R = 6.5\,R_\odot$)
$G_{\rm BP}$	55× (49)	26× (23)	247×	118×
G	29× (28)	14× (13)	100×	48×
$G_{\rm RP}$	18× (18)	8.4× (8.2)	49×	24×
J	8.7× (9.1)	4.1× (4.2)	18×	8.6×
H	6.8× (7.1)	3.2× (3.3)	12×	6.0×
K	5.8× (6.0)	2.7× (2.8)	10×	4.7×

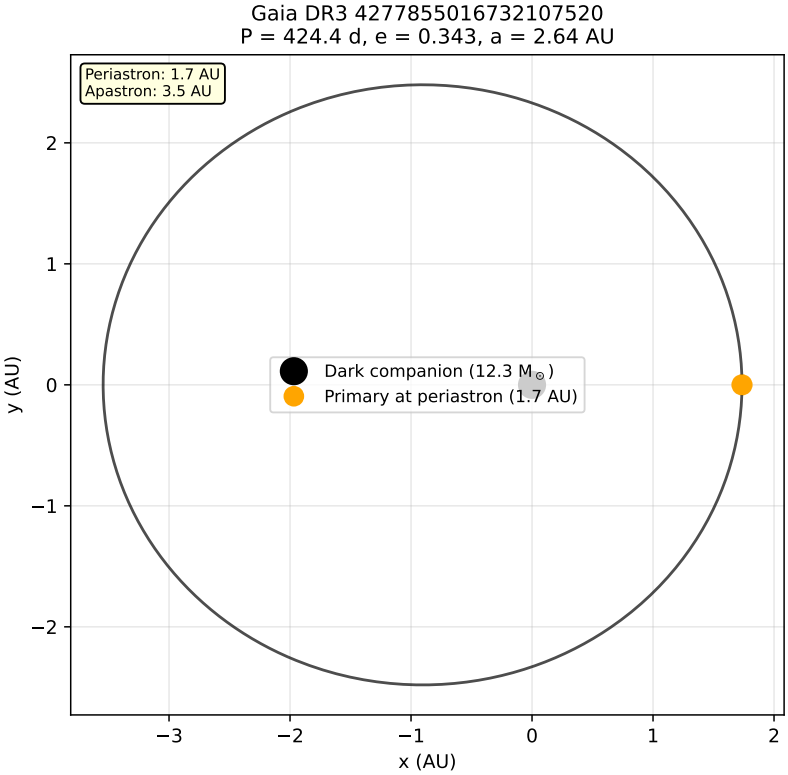

Figure A1. Orbital geometry of the Gaia DR3 4277855016732107520 system. The primary (GSP-Phot classification: evolved; unconfirmed) orbits the unseen companion (black circle at the focus) with period $P = 424.4\,\mathrm{d}$ and eccentricity $e = 0.34$. The periastron distance is 1.73 AU and the apastron distance is 3.55 AU.

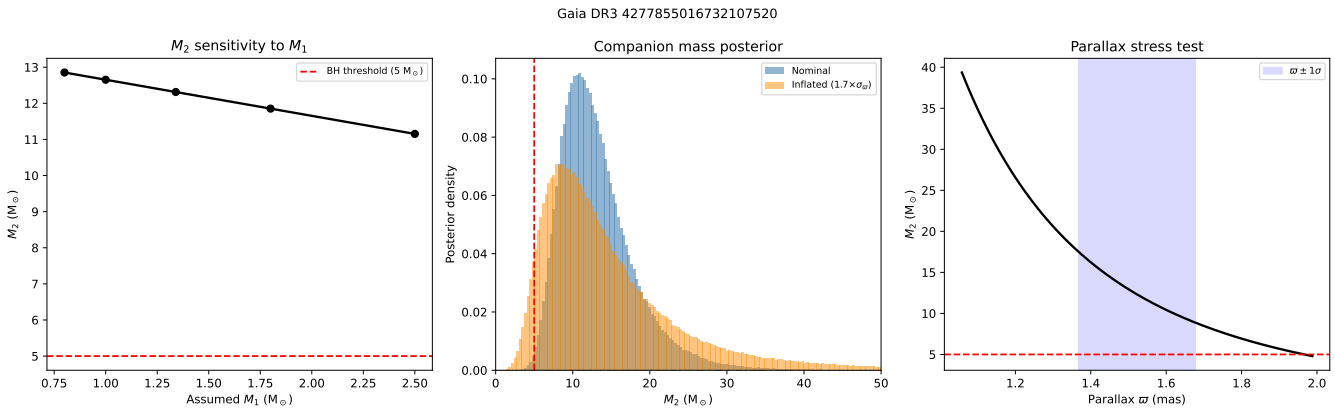

Figure A2. Companion mass posterior from Monte Carlo sampling. *Left:* Sensitivity to primary mass M_1 (all at nominal parallax uncertainty). *Centre:* Posterior with nominal (blue) and inflated ($\times 1.7$; orange) parallax uncertainty. Vertical dashed lines mark the $5\,\mathrm{M}_\odot$ and $10\,\mathrm{M}_\odot$ thresholds. The wide credible intervals reflect the $M_{\mathrm{tot}} \propto \varpi^{-3}$ scaling. *Right:* Companion mass as a function of parallax, illustrating the steep ϖ^{-3} dependence. These posteriors propagate published parameter uncertainties only; they do not account for potential systematic errors in the `Orbital` model itself.

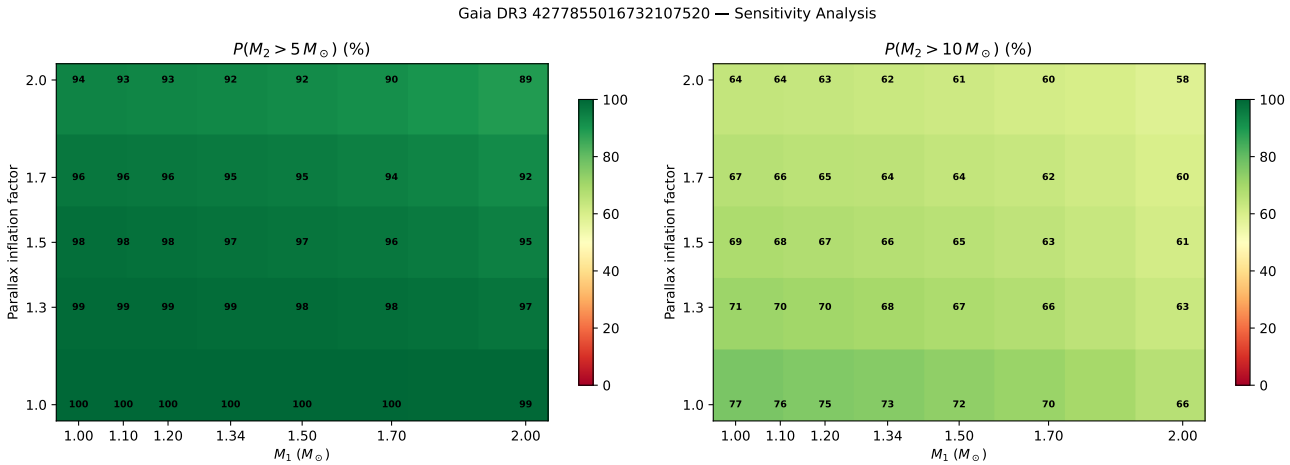

Figure A3. Sensitivity analysis showing $P(M_2 > 5\,\mathrm{M}_\odot)$ (*left*) and $P(M_2 > 10\,\mathrm{M}_\odot)$ (*right*) as a function of adopted primary mass M_1 and parallax-error inflation factor. At the fiducial values ($M_1 = 1.34\,\mathrm{M}_\odot$, inflation $= 1.7$), $P(M_2 > 5\,\mathrm{M}_\odot) \approx 95$ per cent.

Table A3. PARSEC-calibrated primary mass M_1 at the catalogue $T_{\rm eff} = 5922\,{\rm K}$ and $\log g = 2.93$ for selected metallicities and ages (script 13). A Monte Carlo draw over the full parameter space gives median $M_1 = 1.09\,{\rm M}_\odot$ (68% CI [0.77, 1.62]; 90% CI [0.63, 2.12]). Note: the grid input $T_{\rm eff}$ and $\log g$ are the same GSP-Phot values used for the primary-mass estimate in Section 3.3; this cross-check therefore shares the same vulnerability to RUWE-induced bias.

[Fe/H]	M_1 (M$_\odot$) at age (Gyr)				
	1	3	5	7	12
−0.5	2.26	1.40	1.22	1.10	1.00
−0.1	1.97	1.22	1.06	0.96	0.87
0.0	1.90	1.18	1.03	0.92	0.84
+0.3	1.71	1.06	0.93	0.83	0.76

Table A4. Alternative scenario assessment. Verdicts follow a four-tier system (Section 5). *All verdicts are conditional on the Gaia* `Orbital` *solution and adopted primary-star model.*

Scenario	Verdict	Key constraint
MS companion	Excluded*	100–247× flux excess
White dwarf	Excluded*	$M_2 \approx 9\,M_{\rm Ch}$
Neutron star	Excluded*	$M_2 \approx 5\,M_{\rm TOV}$
Triple system	Strongly disfavoured*	SED + stability
Stripped He	Not favoured	NUV deficit 2700×
Artefact	**Requires follow-up**	`Orbital` only
Chance align.	**Requires follow-up**	`Orbital` only

*Conditional on the Gaia orbital solution being correct.

Table A5. Sensitivity of the companion mass posterior to the primary mass and parallax-error inflation factor ($\sigma_\varpi \to f \times \sigma_\varpi$). Each cell reports the median M_2 and $P(M_2 > 5\,\mathrm{M}_\odot)$ from 2×10^5 Monte Carlo draws.

M_1 ($\mathrm{M}_\odot$)	Inflation = 1.0	Inflation = 1.3	Inflation = 1.7	Inflation = 2.0
0.80	12.9 / 99.8%	12.8 / 98.7%	12.9 / 96.0%	12.8 / 93.2%
1.00	12.7 / 99.8%	12.6 / 98.5%	12.6 / 95.7%	12.6 / 92.8%
1.34	12.3 / 99.7%	12.3 / 97.6%	12.3 / 95.4%	12.4 / 91.7%
1.50	12.2 / 99.6%	12.1 / 97.4%	12.1 / 95.0%	12.1 / 91.0%
2.00	11.7 / 99.4%	11.6 / 96.8%	11.6 / 93.7%	11.6 / 89.3%

Table A6. Bright ($G < 12$) Gaia DR3 `Orbital` BH candidates ranked by predicted edge-on radial-velocity semi-amplitude $K_{1,\mathrm{edge}}$ (Equation 6). Sixteen of the 47 `Orbital` BH candidates in the working catalogue satisfy the brightness cut. M_2 is the pipeline-reported companion mass; σ_{NSS} the Gaia significance; "Tier" the catalogue quality flag; "Status" the published vetting state. The target of this paper is highlighted in bold. *Rankings reflect observational accessibility and are not evidence for the reality of any individual solution.*

Rank	Gaia DR3 source ID	G (mag)	P (d)	e	M_2 ($\mathrm{M}_\odot$)	$K_{1,\mathrm{edge}}$ ($\mathrm{km\,s^{-1}}$)	σ_{NSS}	Tier	Hem.	Status
1	**4277855016732107520**	**11.25**	**424**	**0.34**	**12.3**	**65.0**	**75.4**	**Gold**	**Both**	**This work**
2	6281177228434199296	11.26	154	0.18	5.3	62.7	24.3	Gold	Both	Not vetted
3	4507516366984668032	8.89	813	0.51	18.2	48.1	20.3	Gold	Both	Not vetted
4	6042980300715906176	11.96	678	0.22	9.1	47.8	76.3	Gold	Both	Not vetted
5	6033350812290616448	11.75	857	0.41	6.9	41.7	55.5	Gold	Both	Not vetted
6	3153895060707668608	9.34	904	0.00	16.6	41.7	18.1	Silver	Both	Not vetted
7	245948793944575360	11.80	549	0.07	5.0	35.1	48.4	Gold	N	Not vetted
8	6656986282721029120	6.47	444	0.58	5.5	31.6	28.0	Gold	S	Not vetted
9	2946055236704251520	5.27	755	0.32	11.0	30.6	14.6	Silver	Both	Not vetted
10	6886104886536410368	9.46	768	0.48	6.9	30.0	34.9	Gold	Both	Not vetted
11	4113903195790495232	6.81	1233	0.19	7.3	25.9	6.1	Bronze	Both	Not vetted
12	3067772475266826624	11.64	984	0.17	5.7	25.3	20.8	Silver	Both	Not vetted
13	3972084092912522496	8.80	876	0.15	6.1	25.0	21.0	Gold	Both	Not vetted
14	979031266772932736	9.94	877	0.22	5.1	22.5	18.4	Bronze	N	Not vetted
15	470361114339849472	7.18	999	0.52	6.5	21.7	11.2	Silver	N	Not vetted
16	5103955898378211456	6.88	763	0.27	5.0	20.8	22.5	Gold	Both	Not vetted

Table A7. Comparison with confirmed Gaia dormant BH systems (included for contextual reference only). Gaia BH1–3 each have independent multi-epoch RV orbits confirming the companion mass; this source has only a purely astrometric `Orbital` solution with no RV confirmation. The two classes of evidence are not comparable, and the "This work" column reflects conditional inferences under the Gaia pipeline assumptions.

Property	Gaia BH1	Gaia BH2	Gaia BH3	This work
P (d)	185.6	1277	∼4240	424.4
e	0.45	0.52	0.73	0.343
Primary type	G-dwarf	RGB	Giant	GSP-Phot evolved[a]
M_1 ($\mathrm{M}_\odot$)	0.93	1.07	0.76	1.34
M_2 ($\mathrm{M}_\odot$)	9.62	8.94	32.70	∼12.3[b]
Solution type	Astro+RV	Astro+RV	Astro+RV	`Orbital` only
RV confirmation	Yes	Yes	Yes	**No**
d (pc)	480	1160	590	∼657
G (mag)	13.8	12.3	11.6	11.25
Status	Confirmed	Confirmed	Confirmed	**Candidate**

[a] From GSP-Phot only (RUWE = 9.31); no independent spectral classification.
[b] Pipeline-derived mass under Gaia NSS assumptions; 68% CI [7.1, 22.7]; 90% CI [5.0, 35.9] with ×1.7 inflation.